

## 1. Superconducting Qubits & Processor Architecture

The 2026 Hyperion-X quantum processor demonstrates a landmark achievement in superconducting qubit architectures, scaling to 1,024 physical qubits with average coherence times (T1 and T2) surpassing 320 microseconds.

Cross-talk mitigation is achieved through 3D coaxial wiring and tunable couplers, yielding single-qubit gate fidelities of 99.94% and two-qubit CZ gate fidelities of 99.65%. This architecture provides the necessary gate depth to execute deep quantum circuits prior to decoherence.

Thermal dissipation remains under 12 milliwatts at the 15 millikelvin stage of dilution refrigerators, validating scalable cryogenic modular interconnects.

## 2. Surface Code Error Correction Benchmarks

Fault-tolerant quantum computing requires continuous syndrome measurement. Implementing a distance-7 surface code on the Hyperion-X architecture demonstrated a suppressed logical error rate of  $1.2 \times 10^{-6}$  per syndrome cycle.

The physical error threshold remained well below the critical 0.75% limit. Real-time neural decoder chips integrated at the 4 Kelvin stage decoded syndromes in under 450 nanoseconds, outperforming classical FPGA pipelines by 4.2x.

This marks the first empirical proof of exponential error suppression with increasing code distance across superconducting physical lattices.

### 3. Quantum Advantage & Computational Chemistry

Simulations of the FeMo-cofactor nitrogenase active site were completed in 184 seconds on the 1,024-qubit system. Classical tensor-network approximations on the Frontier supercomputer required 4,200 years of equivalent compute.

Energy ground states were determined to within 0.8 kcal/mol chemical accuracy, unlocking novel catalyst pathways for low-temperature synthetic ammonia production.

Future iterations aim to deploy 4,096 physical qubits with photonic quantum repeaters for distributed cluster execution by Q4 2027.